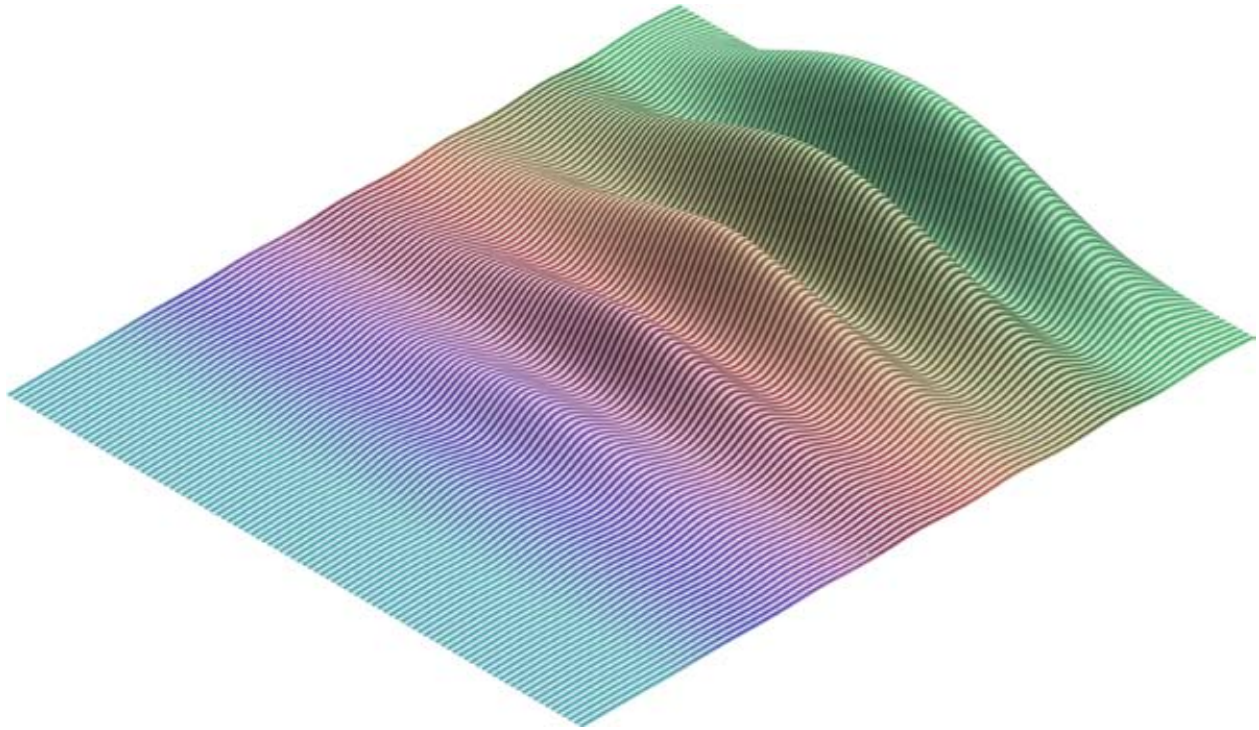




Project10X's
**Semantic Wave 2008 Report:
Industry Roadmap to Web 3.0 &
Multibillion Dollar Market Opportunities**

EXECUTIVE SUMMARY
February 2008

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Project10X's

Semantic Wave 2008 Report:

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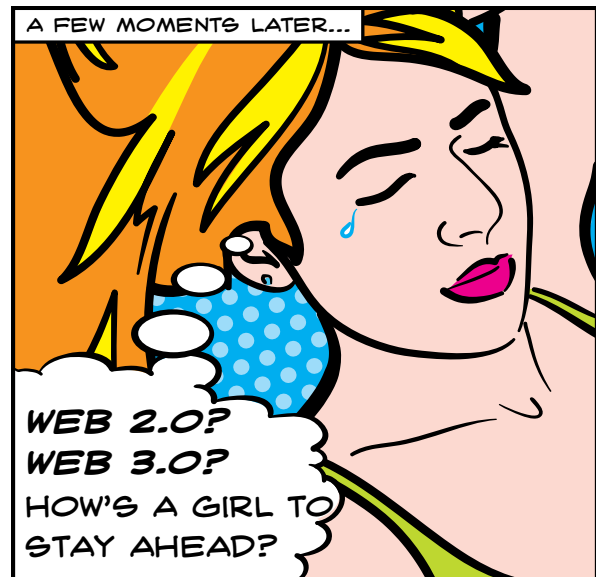
Dear reader,

Project10X is pleased to announce publication of a comprehensive, ground-breaking 400-page study of semantic technologies and their market impact entitled *Semantic Wave 2008: Industry Roadmap to Web 3.0 and Multibillion Dollar Market Opportunities*. This report charts the evolution of the internet from Web 2.0 to Web 3.0, the emergence of semantic technologies for consumer and enterprise applications, and the growth of multi-billion dollar markets for Web 3.0 products and services. It is must reading for investors, technology developers, and enterprises in the public and private sector who want to better understand semantic technologies, the business opportunities they present, and the ways Web 3.0 will change how we use and experience the internet for pleasure and profit. Enjoy this free summary of Project10X's Semantic Wave 2008 Report, and be sure to...[Order your copy of the Semantic Wave 2008 Report. See ordering information on page 27!](#)

Then, if you'd like to take your exploration of these topics a step further, and meet the people and companies that are already bringing semantic technology to life, then you should strongly consider attending [SemTech 2008](#) — the fourth annual conference on Semantic Technology — taking place in San Jose, California on May 18-22, 2008 at the Fairmont Hotel. We'd love to see you at SemTech — I hope you can make it.



Mills Davis
Washington, DC USA



What is the semantic wave?

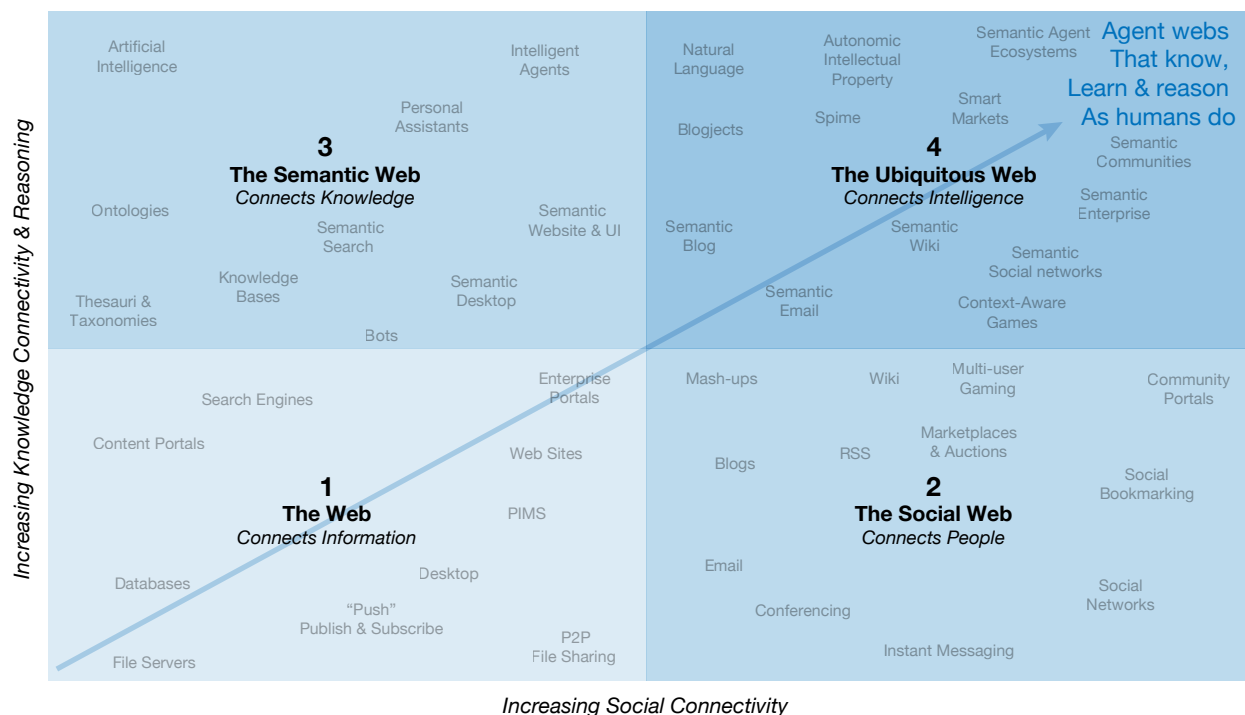
A tidal wave of four Internet growth stages.

The semantic wave embraces four stages of internet growth. The first stage, *Web 1.0*, was about connecting information and getting on the net. *Web 2.0* is about connecting people — putting the “I” in user interface, and the “we” into Webs of social participation. The next stage, *Web 3.0*, is starting now. It is about representing meanings, connecting knowledge, and putting these to work in ways that make our experience of internet more relevant, useful, and enjoyable. *Web 4.0* will come later. It is about connecting intelligences in a ubiquitous Web where both people and things reason and communicate together.

Project10X's Semantic Wave 2008 Report tells the story of Web 3.0. Over the next decade, Web 3.0 will spawn multi-billion dollar technology markets that will drive trillion dollar global economic expansions to transform industries as well as our experience of the internet. The Semantic Wave 2008 report examines drivers and market forces for adoption of semantic technologies in Web 3.0 and maps opportunities for investors, technology developers, and public and private enterprises.

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Below:
What is the Evolution of the Internet to 2020?



Source: Nova Spivak, Radar Networks; John Breslin, DERI; & Mills Davis, Project10X

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How is Web 3.0 different from previous stages of internet evolution?

Knowledge computing drives new value creation and solves problems of scale and complexity.

The basic shift occurring in Web 3.0 is from information-centric to knowledge-centric patterns of computing. Web 3.0 will enable people and machines to connect, evolve, share, and use knowledge on an unprecedented scale and in new ways that make our experience of the internet better.

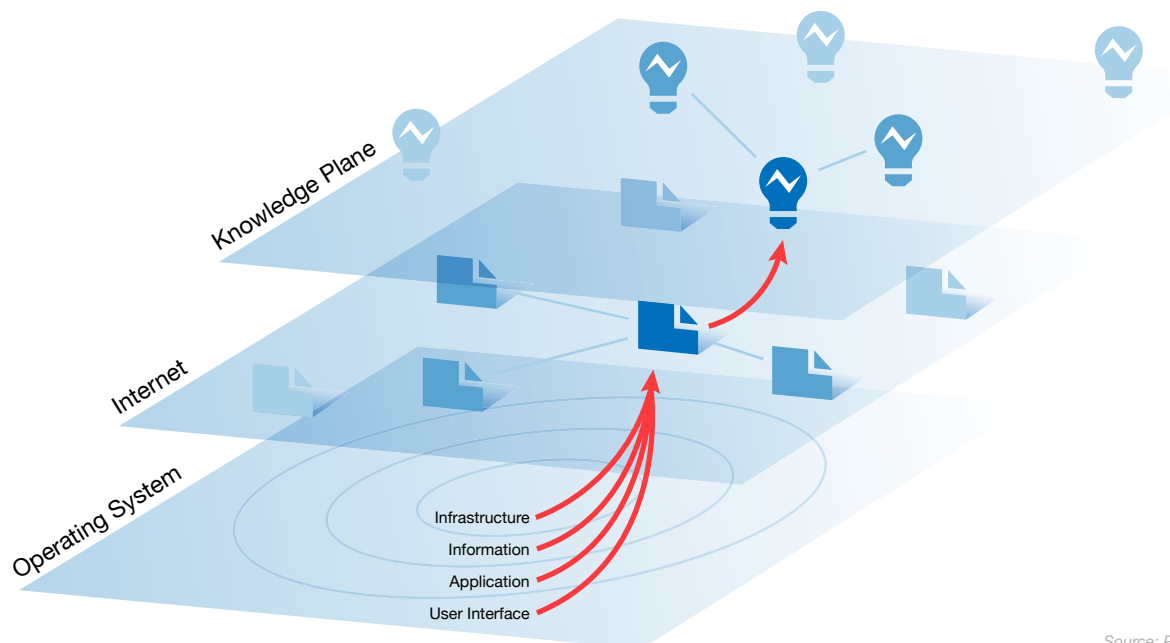
Web growth continues to accelerate. Dimensions of net expansion include communications bandwidth, numbers of people connected, numbers and kinds of devices that are IP-aware, numbers of systems and applications, quantities of information, and types of media. As the internet expands, needs world-wide are outstripping the capacities and capabilities of current information and com-

munications technologies (ICT) and architectures. Information-centric patterns of computing have reached the limit of what they can provide to cope with problems of scale, complexity, security, mobility, rich media interaction, and autonomic behavior.

Web 3.0 will solve these problems and lay a foundation for the coming ubiquitous Web of connected intelligences. The Web 3.0 solution, simply put, is to give the internet a knowledge space. In the following topics we identify key characteristics of this knowledge space, sketch out how its semantic computing works, and examine how Web 3.0 knowledge-centric patterns of computing drive new value creation.

Below:

Web 3.0 — The Internet Grows a Knowledge Plane



Source: Project10X

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What semantic technologies will power Web 3.0?

Digital tools that represent and reason about meanings, theories, and know-how separately from documents, data, and program code.

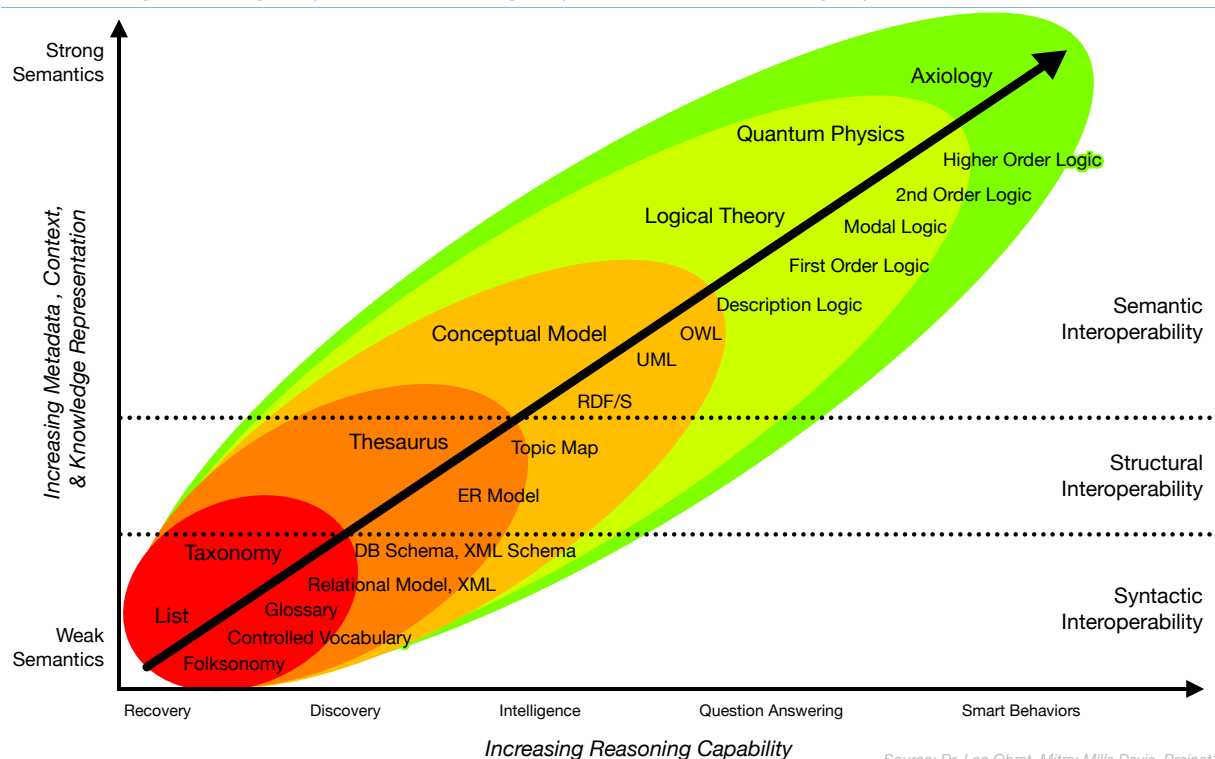
The key notion of semantic technology is to represent meanings and knowledge (e.g., knowledge of something, knowledge about something, and knowledge how to do something, etc.) separately from content or behavior artifacts, in a digital form that both people and machines can access and interpret. As a platform, Web 3.0 will embrace all semantic technologies and open standards that can be applied on top of the current Web. It is not restricted just to current Semantic Web standards.

Web 3.0 will encompass a broad range of knowledge representation and reasoning capabilities including pattern detection, deep linguistics, ontology and model based inferencing, analogy and reasoning with uncertainties, conflicts, causality,

and values. The figure below depicts a spectrum of progressively more capable forms of knowledge representation that spans tag collections (or folksonomies); to dictionaries, taxonomies and thesauri; to schemas and conceptual models; to ontologies and theory-based logics, to axiologies (value-based reasoning), and entirely new uses barely tapped. Reasoning requires knowledge representation. We choose more powerful forms of representation to enable more powerful kinds of reasoning and problem solving. The integration of social Web and semantic technologies in Web 3.0 allows new synergy that lowers the cost of data and knowledge creation, and raises the computational value of gathering it.

Below:

From Searching to Knowing — Spectrum of Knowledge Representation and Reasoning Capabilities



How will Web 3.0 systems connect data, services and applications?

First, they'll integrate knowledge about these applications, content sources, and process flows. Then they'll execute it.

In order to connect systems, integrate information, and make processes interoperable, the first step is to integrate the knowledge about these systems, content sources, and process flows. Today, people do this offline, manually. This approach does not scale. In Web 3.0 both people and applications will connect knowledge in real time using automated and semi-automated methods. Web 3.0 approaches will scale.

Semantically modeled, machine executable knowledge lets us connect information about people, events, locations, times — in fact, any concept that we want to — across different content sources and application processes. Instead of disparate data and applications on the Web, we get a Web of interrelated data and interoperable applications. Recombinant knowledge is represented as concepts, relationships and theories that are sharable and language neutral. Semantic technologies provide the means to unlock knowledge from localized environments, data stores, and proprietary formats so that resources can be readily accessed, shared, and combined across the Web.

In today's Web, each device has an operating system (OS) that provides walled access to its content through a hierarchical file system. Limitations of OS platforms are spurring development of semantic desktops to provide meaning-based, concept-level search, navigation, and integration across varied content sources and applications found on PCs and other devices.

Applications running on OS platforms provide access to the information they have knowledge of, but do not combine easily with others, unless such link-ups have been planned and agreed to in advance by developers. The need to overcome

these limitations of OS platforms including the need for human labor to research and code interfaces is fueling interest in:

Web-tops — platforms spanning multiple OSs connected over the internet,

Mash-ups — two or more data sources or works combined to become a new data source or work,

Context-aware mobility — dynamic composition and personalization of services across devices, networks, locations, and user circumstances, and

Semantic service-oriented architectures — using machine-interpretable descriptions of policies and services to automate discovery, negotiation, adaptation, composition, invocation, and monitoring of Web services.

In Web 3.0, these sorts of capabilities will become intrinsic features of the knowledge space's semantic fabric, and no longer mere one-off hacks or the result of mutually exclusive platform and service plays.

Where do the shared meanings and knowledge in Web 3.0 come from?

From both people and machines. And, to start with, from the Web itself.

Knowledge exists in many forms in today's Web. All computing processes represent some type of knowledge in some way in order to process information, for example: knowledge about how information is organized in order to search it; rules that tell a computer program how to make a decision; or action steps to take to complete a task.

The problem is that existing knowledge on the Web is fragmented and difficult to connect. It is locked in data silos and operating system file system formats. Knowledge is hidden in object-oriented black boxes and layers of stack architecture. It is embedded in program code and squirreled away in proprietary algorithms.

Web 3.0 changes this. The convergence of pattern discovery, deep linguistics, and ontological symbolic reasoning technologies make it feasible to automatically extract embedded and intrinsic knowledge from today's Web. Evolution of semantic social computing will enable communities to create, curate, and share knowledge in human readable and machine executable forms.

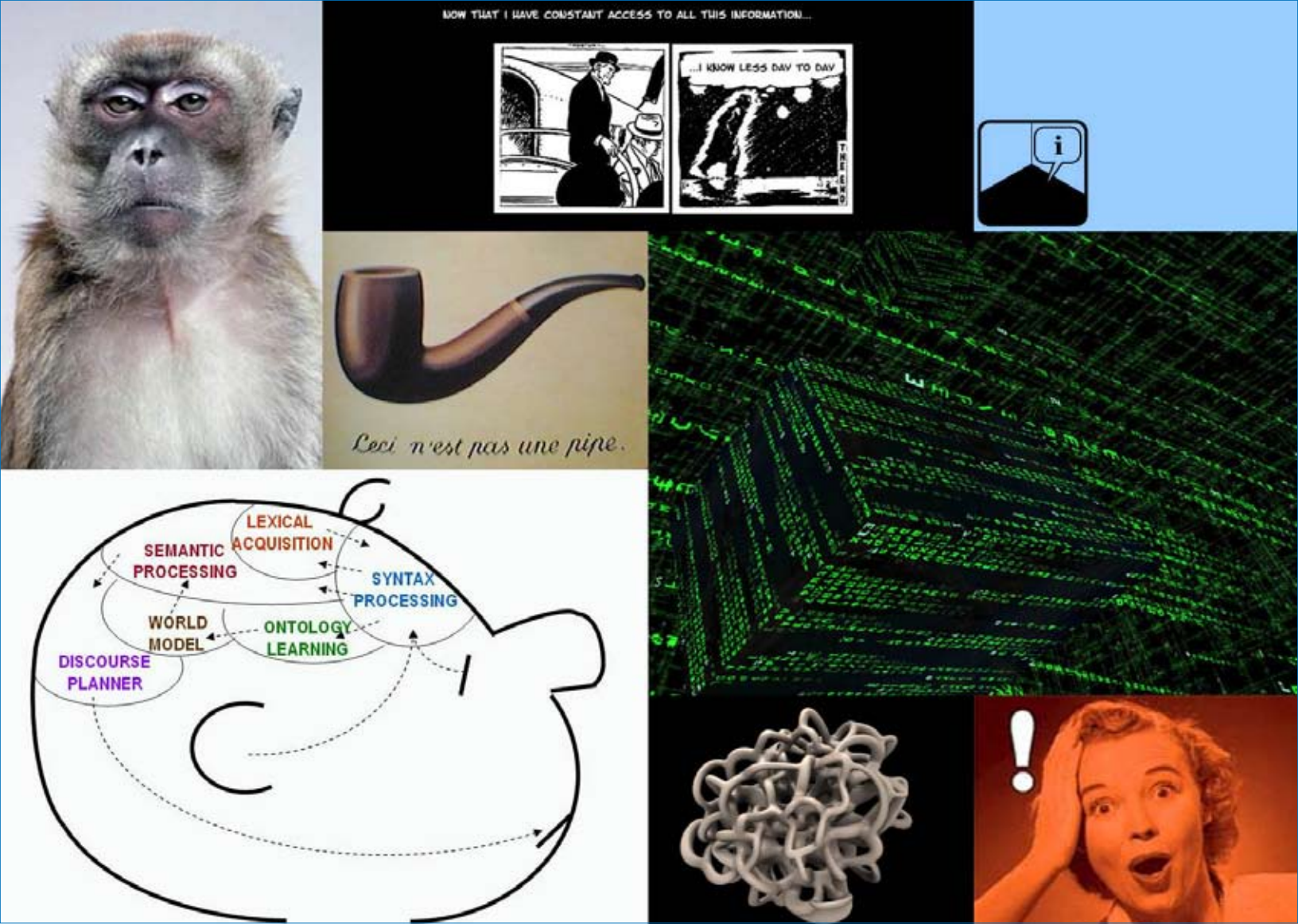
The diagram below contrasts knowledge-centric and information-centric patterns of computing. In Web 3.0, end-user development will increase as computers help generate intelligent services and manage application functionality, security, versioning and changes autonomically.

Below:
What Are Knowledge-centric Patterns of Computing?

Pattern	Information-centric	Knowledge-centric
Who develops software behaviors, knowledge structures, and content?	Producers and enterprises are developers.	Prosumers (consumers) and peer-to-peer producers (groups, communities) do it themselves.
How are different expressions of knowledge handled?	Separate technologies for documents (data, content), models, and behaviors. Closed semantics, hardwired.	Unified platforms handle documents, models & behaviors interchangeably, including pictures & natural language. Massive open local semantics, available everywhere.
Where do knowledge & logic in the system come from?	At design time, from people. At new release, from people. No run-time learning.	At design time, from people. At run time, from user input and from system learning.
What are the patterns for system learning?	No system learning. No autonomies. New knowledge requires new version of code.	System learns and evolves from use by people. Machine observes & learns from environment. Autonomics — self* learning and adaptation.
What are the patterns for knowledge representation and computation?	Process-centric, cycle time intensive. Directional algorithms and procedures. Embedded knowledge — logic, structure locked in code. Relational operators. First-order logic.	Data-centric, storage-intensive. Semantic operators. Sequence neutral graph reasoning. External declarative knowledge structures. Semantic and value-based reasoning with full spectrum of logic.
What are the patterns for underlying infrastructure?	Predefined configurations. Black-box objects. Stacks. Single processors. Local stores.	Adaptive, self-optimizing configurations. Ubiquitous semantic Webs, meshes & grids. Transparent semantic agents. Multi-core, multi-threaded processors. Federated stores and processes. Semantic ecosystems and social autopoiesis (self-organization, planning, etc.).
What are the patterns for security?	Separate role-based security for each system. Black boxes, lack of transparency, and human intervention make network security problematic.	Autonomic identity and security with concept level granularity across all IP entities, relationships, services, etc. Building block transparency = security by design.
What are patterns for versioning and change management?	Manual change management and versioning. Human architected. Central planning. Brittle.	Automated change management & versioning. Autonomic intellectual property, emergent behaviors, self-managed. Robust.

Source: Project10X

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What new capabilities will Web 3.0 knowledge-centric computing enable?

Systems that know, learn, and can reason as humans do.

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When knowledge is encoded in a semantic form, it becomes transparent and accessible at any time to a variety of reasoning engines.

Previously, if knowledge was in a document or set of documents, then it was fixed when published in a form only humans could read. Or, if knowledge was encoded in a computer program, then it was opaque and hidden in objects or in procedures that were fixed at design time, and hence a "black box" so that the logic is not visible to any other process that had not been pre-programmed with common knowledge.

In Web 3.0, knowledge lives, evolves and is stored transparently (as "glass boxes"). It can be used, validated, added to, combined with other knowledge at run time by multiple systems. This enables a system to "learn" to do things that the system designer did not anticipate. This is an important shift from IT as it has been practiced until now.

Web 3.0 systems will be designed so that they get better with use and scale. Their architectures will enable learning. One way is that their users can evolve them by adding knowledge and capabilities to them. Another way is that systems may learn by themselves how to respond to changes in their environments.

How will Web 3.0 overcome the fragmentation of information, processes, and application functionality?

By interrelating the myriad forms of language that people and machines used to encode thoughts, share meanings, and connect knowledge.

Until now, knowledge on the Web has been expressed in separate forms such as documents, imagery, patterns, structural models, and program code. Computers that produced these artifacts mostly have been used as electronic pencils, with little (if any) understanding of what the writing meant, and no ability to interpret other ways of expressing the same idea (such as through graphics, images, video, computer languages, formal languages, and other natural languages, etc.).

In Web 3.0, the myriad forms of language in which knowledge is expressed begin to get interrelated, connected, and made interchangeable with each other, for example: combining knowledge from one or more sources, or from one or more formats, or from one time and place with other contexts.

To illustrate, policies are typically written out as documents. But, this same knowledge can be modeled as a data structure or as decision rules. Also, policies can be hard coded into software objects and procedures. Using semantic technologies we can represent, connect, and manage the

knowledge from all of these different forms at the level of concepts, and maintain each artifact. This sort of “transemanic” or multi-lingual capability leads to computer systems that can:

- Capture knowledge from different sources such as sensors, documents, pictures, graphics, and other data and knowledge resources,
- Interpret and interrelate these different ways of expressing ideas with each other,
- Share what they know with people and machines, and
- Re-express, and communicate what they know in different contexts, information formats, and media.

Below:
How Do Humans Encode Thoughts and Share Knowledge and Meaning?

Natural language	Documents, speech, stories
Visual language	Tables, graphics, charts, maps, illustrations, images
Formal language	Models, schema, logic, mathematics, professional and scientific notations
Behavior language	Software code, declarative specifications, functions, algorithms
Sensory language	User experience, human-computer interface

Source: Project10X

How does Web 3.0 tap new sources of value?

By modeling knowledge, adding intelligence, and enabling learning.

The value drivers for Web 3.0 are huge. The table below highlights five categories of challenges, semantic capabilities that address these needs, and the value drivers associated with these semantic capabilities. Semantic technologies have the potential to drive 2-3 order of magnitude improvements in capabilities and life cycle economics through cost reductions, improved efficiencies, gains in effectiveness, and new functionalities that were not possible or economically feasible before now. New sources of value include:

1. Value from knowledge modeling — Semantic models are sharable, recombinant, and executable. To model first, then execute the knowledge reduces time, risk, and cost to develop and evolve services and capabilities. Semantic model-based approaches achieve added development economies through use of (a) shared knowledge models

as building blocks, (b) autonomic software techniques (goal-oriented software with self-diagnostic and self-management capabilities such as self-configuration, self-adaptation, self-optimization, etc.), and (c) end-user and do-it-yourself life-cycle development methodologies (rather than requiring intervention by IT professionals). Knowledge that is sharable, revisable, and executable is key applications where facts, concepts, circumstances, and context are changing and dynamic.

2. Value from adding intelligence — A working definition of intelligence is the ability to acquire, through experience, knowledge and models of the world (including other entities and self), and use them productively to solve novel problems and deal successfully with unanticipated circumstances. A key new source of value is adding intelligence to the user interface, to applications,

Below:
How Do Semantic Technologies Drive Value?

Challenges	Semantic Capabilities	Value Drivers
1. Development: Complexity, labor-intensity, solution time, cost, risk	Semantic automation of “business need-to-capability-to-simulate-to-test-to-deploy-to-execute” development paradigm	Semantic modeling is business rather than IT centric, flexible, less resource intense, and handles complex development faster.
2. Infrastructure: Net-centricity, scalability; resource, device, system, information source, communication intensity	Semantic enablement and orchestration of transport, storage, and computing resources; IPv6, SOA, WS, BPM, EAI, EII, Grid, P2P, security, mobility, system-of-systems	In the semantic wave, infrastructure scale, complexity, and security become unmanageable without semantic solutions.
3. Information: Semantic interoperability of information formats, sources, processes, and standards; search relevance, use context	Composite applications (information & applications in context powered by semantic models), semantic search, semantic collaboration, semantic portals	Semantic interoperability, semantic search, semantic social computing, and composite applications & collaborative knowledge management become “killer apps.”
4. Knowledge: Knowledge automation, complex reasoning, knowledge commerce	Executable domain knowledge-enabled authoring, research, simulation, science, design, logistics, engineering, virtual manufacturing, policy and decision support	Executable knowledge assets enable new concepts of operation, super-productive knowledge work, enterprise knowledge superiority, and new intellectual property.
5. Behavior: Systems that know what they’re doing	Robust adaptive, autonomic, autonomous system behaviors, cognitive agents, robots, games, devices, and systems that know, learn, and reason as humans do	Semantic wave systems learn and reason as humans do, using large knowledgebases, and reasoning with uncertainty and values as well as logic.

and to infrastructure. An intelligent system or agent is a software program that learns, cooperates, and acts autonomously. It is autonomic and capable of flexible, purposeful reasoning action in pursuit of one or more goals. An intelligent user interface (UI) knows about a variety of things such as system functionality, tasks users might want to do, ways information might be presented or provisioned. Intelligent UIs know about the user (via user models), which enables tailoring system behavior and communications. Adding intelligence helps users perform tasks, while making working with the computer more helpful, and as invisible as possible. As a result, systems do more for the user, yield more relevant results with less effort, provide more helpful information and interaction, and deliver a more enjoyable user experience. Adding intelligence can produce ten-fold gains in communication effectiveness, service delivery, user productivity, and user satisfaction.

3. Value from learning — Machine learning is the ability of computers to acquire new knowledge from past cases, experience, exploration, and user input. Systems that learn increase in value during their lifetime. Their performance improves. They get better with use, and with scale. In addition to new or improved capabilities, systems that learn during operation may improve system life cycle economics by (a) requiring less frequent upgrading or replacement of core software components, and (b) enabling new incremental extensions to revenue models through add-on knowledgeware and software-as-a-service.

4. Value from semantic ecosystem — An ecosystem is a self-sustaining system whose members benefit from each other's participation via symbiotic relationships (positive sum relationships). Principle drivers for semantic infrastructure and ecosystem include the economics of mobility, scale, complexity, security, interoperability, and dynamic change across networks, systems, and information sources. These problems are intrac-

table at Web scale without semantics. The corollary is the need to minimize human labor needed to build, configure, and maintain ultra-scale, dynamic infrastructure.

Semantic ecosystems that emerge in Web 3.0 will consist of dynamic, evolve-able systems consisting of ensembles (societies) of smart artifacts. This means a shift in design focus from static, performance-driven design to: (a) design for robustness & resilience; (b) design for uncertainties; (c) design for distributed, autonomous pervasive adaptation; (d) design for organically growing systems; and (e) design for creating self-evolving services.

Current systems including the internet are designed to operate with predefined parameters. Change spells trouble. Mobility is a problem. Semantic ecosystems will be future-proof, able to grow dynamically, evolve, adapt, self-organize, and self-protect. Web 3.0 will lay the foundations for ubiquitous Web including autonomic intellectual property, Web-scale security and identity management, and global micro-commerce in knowledge-based assets. The value vector for semantic infrastructure is 2-4 orders of magnitude gains in capability, performance, and life cycle economics at Web scale.

Semantic Wave Technology Trends

The Semantic Wave 2008 Report examines over 100 application categories & more than 270 companies pursuing semantic products and services.

A broad range of semantic technologies will power Web 3.0. The technology section of Project10X's Semantic Wave 2008 Report examines Web 3.0 technology themes from multiple perspectives. It shows how innovations in each area will drive development of new categories of products, services, and solution capabilities. Technology perspectives include:

Semantic user experience — concerns how I experience things, demands on my attention, my personal values.

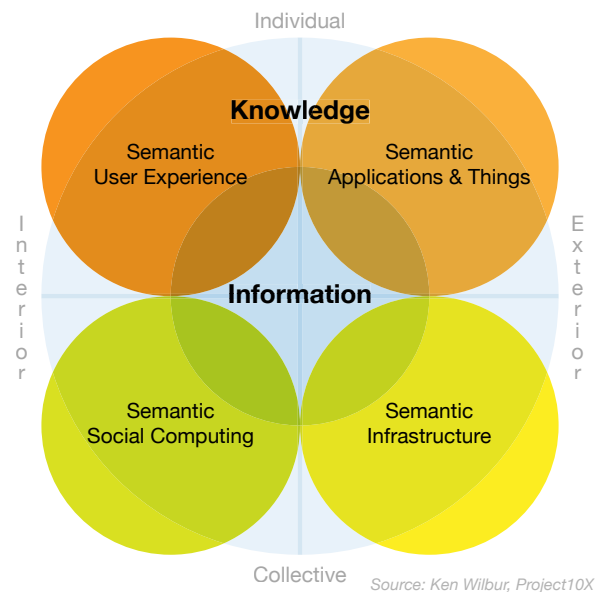
Semantic social computing — concerns our lived culture, intersubjective shared values, & how we collaborate and communicate.

Semantic applications, and things — concerns objective things such as product structure & behavior viewed empirically.

Semantic infrastructure — concerns interobjective network-centric systems and ecosystems.

Semantic development — concerns Webs of meanings, systems that know and can share what they know, and architectures of learning, which make semantic solutions different.

Semantic Wave 2008 spotlights trends in each of these areas and examines role of semantic technologies in over 100 application categories. An addendum to the report surveys more than 270 companies that are currently researching and developing semantic wave technology products and services.



Above:
Semantic Technology Perspectives

Technology Trend 1—Semantic User Experience

Intelligent user interfaces drive gains in user productivity & satisfaction.

The Semantic Wave 2008 Report explores the impact of semantic technologies on user experience. User experience is the sum of interactions and overall satisfaction that a person has when using a product or system. Semantic user experience is the addition intelligence and context-awareness to make the user interface more adaptive, dynamic, advisory, proactive, autonomic, and autonomous, and the resulting experience easier, more useful, and more enjoyable.

Attention is the limited resource on the internet — not disk capacity, processor speed or bandwidth. Values shape user experience. Simplicity, versatility and pleasurability are the new watchwords. Context is king. Mobility, wireless, and video are the new desktop. Seamless services anytime, any where. Users are prosumers, creating content, participating in peer production, taking control of consumption. Trends in user interface (UI) are towards personal avatars; context-aware, immersive 3D interaction; and reality browsing, and augmented reality.

Identity is information used to prove the individuality of a person as a persisting entity. The trend is towards semantic avatars that enable individuals to manage and control their personal information, where ever it is across the net. Context is information that characterizes a situation of an entity,

person, object, event, etc. Context-awareness is using this knowledge to sense, predict, interpret, and respond to a situation.

Web 3.0 browsers will understand semantics of data, will broker information, and automatically interpret metadata. The emerging display landscape (depicted above) will be semantically connected and contextually aware. It will unify displaying and interacting, and will personalize experience. Reality browsing is querying the physical world live and up close from anywhere. Augmented reality is bringing the power of the Web to the point of decision, by combining real world and computer generated data. Semantic rich internet applications will exploit higher bandwidth content dimensionality, context sensitivity, and expanded reasoning power for dynamic visualization and interaction in the UI.

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Technology Trend 2 — Semantic Social Computing

Collective knowledge systems become the next “killer app.”

The Semantic Wave 2008 Report explores the role of semantic technologies in the evolution of social computing. Social computing is software and services that support group interaction. Semantic social computing adds an underlying knowledge representation to data, processes, services, and software functionality.

Semantic technologies will enrich many categories social applications including instant messaging, e-mail, bookmarking, blogging, social networking, wikis, user driven “communitainment”, and do-it-yourself applications and services. For example, semantic technologies will enable social computing applications to provide concept-based rather than language-based search and navigation across most standard applications, document types, and file formats, regardless where these resources reside on the net, be it a desktop, mobile device or server, etc.

A key trend in Web 3.0 is toward collective knowledge systems where users collaborate to add content, semantics, models, and behaviors, and where systems learn and get better with use. Collective knowledge systems combine the strengths of social Web participation with semantic Web

integration of structure from many sources. Key features of Web 3.0 social computing environments include (a) user generated content, (b) human-machine synergy; (c) increasing returns with scale; and (d) emergent knowledge. Incorporating new knowledge as the system runs is what enables Web 3.0 systems to get smarter.

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Technology Trend 3 — Semantic Applications

New capabilities, concepts of operation, & improved life cycle economics.

The Semantic Wave 2008 Report examines the emerging role of semantic technologies in more than 100 consumer and enterprise application categories. Semantic applications put knowledge to work. Areas covered in the report include: (a) semantics in commercial off the shelf software such as ERP, CRM, SCM, PLM, and HR; (b) ontology-driven discovery in law, medicine, science, defense, intelligence, research, investigation, and real-time document analysis; (c) risk, compliance and policy-driven processes such as situation assessment, exceptions, fraud, case management, and emergency response; (d) knowledge-intensive processes such as modeling & simulation, acquisition, design, engineering, and virtual manufacturing; (e) network & process management such as diagnostics, logistics, planning, scheduling, security, and event-driven processes; (f) adaptive, autonomic, & autonomous processes such as robotics and intelligent systems; and (g) systems that know, learn & reason as people do such as e-learning, tutors, advisors, cognitive agents, and games

Key trends toward semantic applications are:

From knowledge in paper documents, to digital documents, to knowledge (semantic models), to semantic agents;

From static and passive functional processes,
to active, adaptive, and dynamic processes,
to autonomic to autonomous processes;

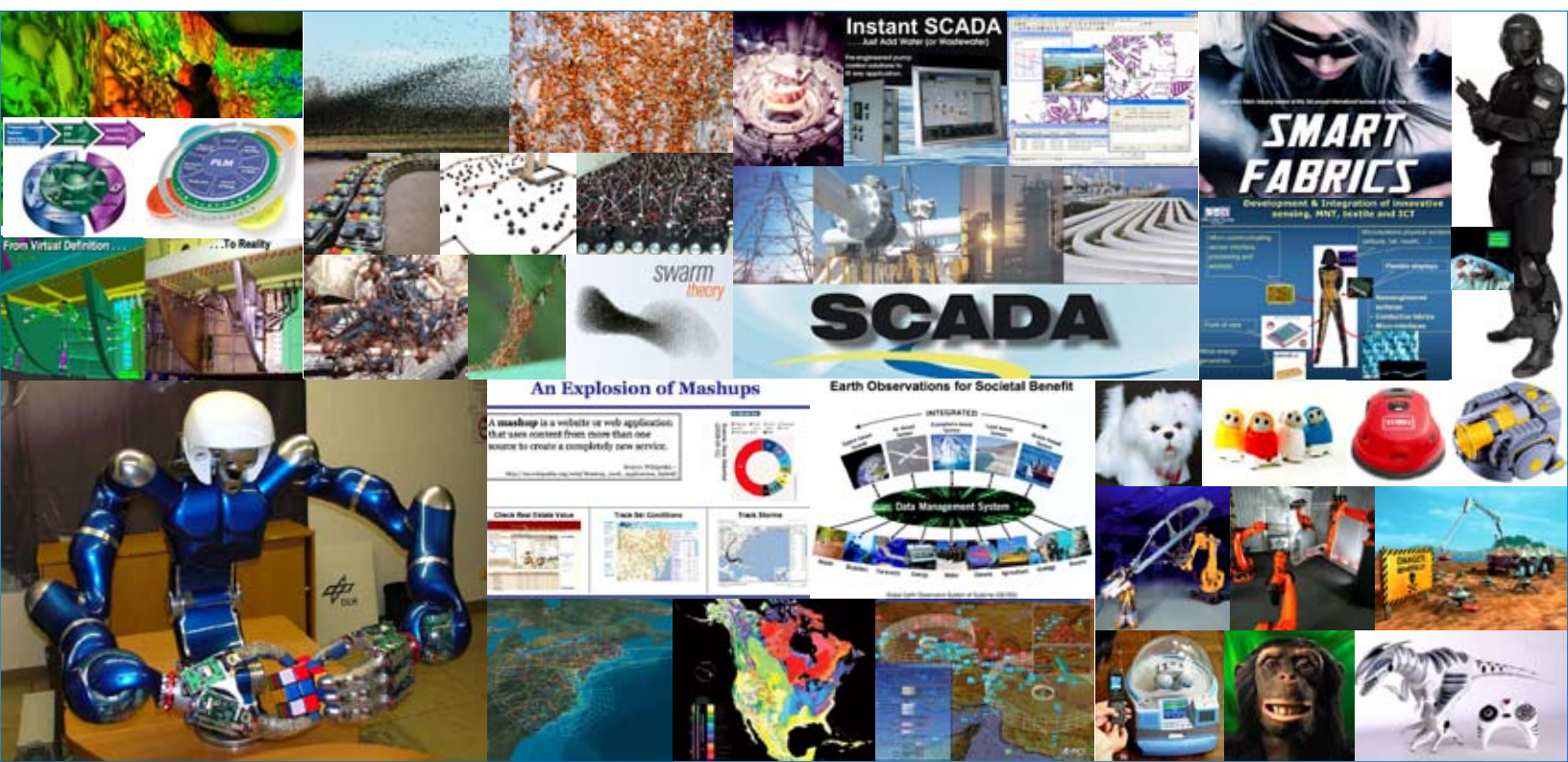
From programmer encoded interpretations of meaning and logic at design time, to computer interpretation of meaning and logic at run time;

From smart program code to smart data;

From search to knowing; and

From reasoning with SQL to first order logic, to complex reasoning with uncertainty, conflict, causality, and values for the purposes of discovery, analysis, design, simulation, and decision-making.

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Technology Trend 4 — Semantic Infrastructure

A knowledge space solves problems of scale, complexity and security.

The Semantic Wave 2008 Report examines the role of semantic technologies in infrastructure. Infrastructure is the basic features of a system such as networks, facilities, services, and installations that are needed for the functioning of internet-based communities. By adding a knowledge dimension to this underlying structure, semantic infrastructures provide solutions to problems of Integration, interoperability, parallelism, mobility, ubiquity/pervasiveness, scale, complexity, speed, power, cost, performance, autonomies, automation, intelligence, identity, security, ease of programming, and ease of use.

Information and communications technology (ICT) has reached the limits of what it can do with stack architecture, object orientation, first-order logic, and fixed, embedded knowledge (i.e., in code) with no learning, or with architected development versus emergent solutions. Semantic technologies provide the first path forward to overcome the limitations of these existing approaches.

Trends toward semantic infrastructure include:

Computing diverges into declarative (brain) and procedural (sensory organs) lines of development.

Storage moves from flat files, to centralized “bases” with relational operators, to federated “spaces” with native semantic operators. The trend is toward high-performance semantic processing at scale and representations that support nearly unlimited forms of reasoning.

Transport moves from dial-up, to broad band, to video bandwidth. Mobility is the new platform, and semantic technologies are needed to deliver seamless, customizable, context aware services, any time, any where.

Processor technology goes parallel, multi-core, multi-threaded, and specialized.

Displays become a landscape of interoperable devices of differing characteristics, sizes and capabilities. Boundaries between virtual and real dissolve in planned and unplanned ways. The trend is towards immersive experience and reality browsing.

Longer term, the trend is towards every thing becoming connected, somewhat intelligent, somewhat self-aware, socially autopoietic, and autonomically capable of solving problems of complexity, scale, security, trust, and change management.

Technology Trend 5 — Semantic Development

Semantic modeling reduces time, risk, and cost to develop solutions.

The Semantic Wave 2008 Report explores trends in methodology and practices for semantic software and solution development.

A development life cycle is a conceptual model used in project management that describes the stages involved in a system or application development project, typically involving a discovery, feasibility, and planning stage through maintenance of the completed application. Conventional development methodologies include the waterfall model; rapid application development (RAD); the fountain model; the spiral model; build and fix; and synchronize-and-stabilize, etc.

Semantic solution development departs from conventional development. It deals with: (a) Webs of meanings and knowledge from diverse provenance, (b) systems that know and can share what they know, and (c) architectures of learning.

Semantic solutions emerge from a techno-social collaboration that also supports do-it-yourself development. The process is business and user driven versus IT and developer driven. The collective knowledge developed is both human and machine interpretable. Some different skills are required including domain experts, semantic modelers, and semantic user experience designers.

Knowledge is extracted and modeled separately from documents, schemas, or program code so it can be managed across these different forms of expression, shared between applications, aligned and harmonized across boundaries, and evolved. For example, requirements, policies, and solution patterns are expressed as semantic models that execute directly and can be updated with new knowledge as the application runs.

The semantic solution development process is model-driven and knowledge-centric and rather than procedural and document based. Semantic solutions may have zero code. Build cycles are fast, iterative, non-invasive. Semantic solution development typically entails less time, cost, and risk to deploy, maintain, and upgrade.



Semantic Wave Markets

The Semantic Wave 2008 Report sizes markets and presents 150 case studies in 14 horizontal and vertical market sectors.

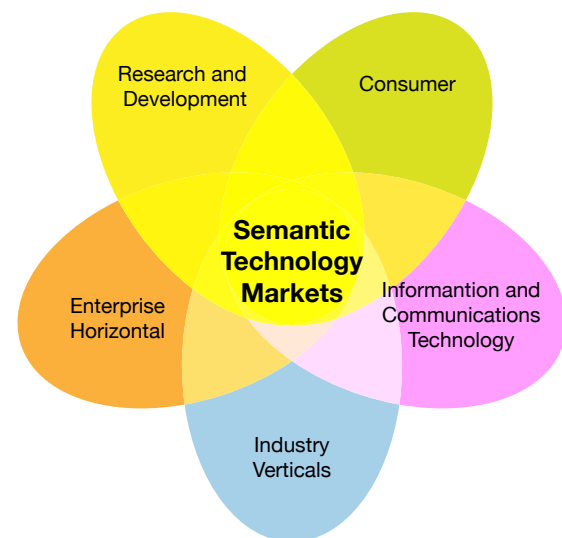
The market section of Project10X's Semantic Wave 2008 Report examines the growth of supply and demand for products, services and solutions based on semantic technologies. Specifically, the report segments and discusses semantic wave markets from horizontal and vertical perspectives:

Horizontal market sectors include: Research and development; Information and communication technologies; Consumer internet; and Enterprise horizontal.

Vertical market sectors include: Advertising, content, entertainment; Defense, intelligence, security; Civilian agencies, state & local government; Education, training; Energy, utilities; Financial services; Health, medical, pharma, life sciences; Information & communications technology; Manufacturing; Professional services; Transportation, logistics; and other services.

Horizontal and vertical market sectors each present multi-billion dollar opportunities in the near-to mid-term. The study sizes markets. It presents 150 case studies in 14 horizontal and vertical sectors that illustrate the scope of current market adoption.

Semantic technologies are spreading out and penetrating into all areas of information and communications technology, all economic sectors, and most categories of application. There are powerful economic drivers. Development and adoption is already global in scope. Market momentum is building. The sweet spot for cross-over and market acceleration is only about a year out.



Source: Project10X

[Above:](#)
[What Are Semantic Wave Markets?](#)

Market Trend 1 — Research & Development

Semantic technologies are a significant and growing focus in global R&D.

The maturation of R&D investments made in the public and private sectors over the past decade is one reason why semantic technologies and Web 3.0 are entering mainstream markets. The diagram below highlights semantic technology areas which are receiving international R&D funding estimated to be more than \$2B per year through the end of the decade.

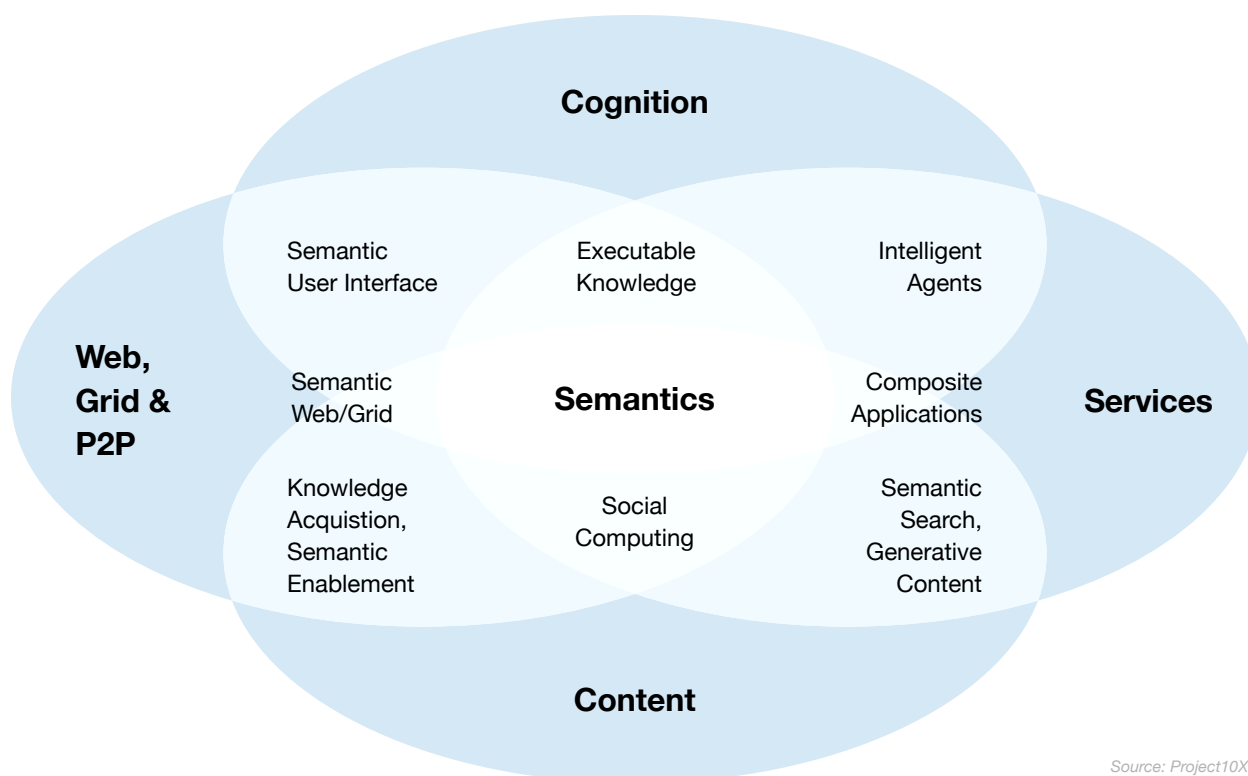
Public sector investment has been significant and is growing in North America, Europe, and Asia. Countries recognize the strategic importance of semantic technologies in the emerging global knowledge economy and are seeking competitive advantage through public sector investments. Historically, it is worth noting that public sector investment to develop ICT technologies has a strong track record, having spawned \$-billion industries repeatedly over the past 40 years.

The Semantic Wave 2008 Report provides summarized examples of public sector R&D programs from organizations such as: DARPA, Air Force Research Laboratories, NASA, and NSF.

Private sector firms accelerate semantic technology R&D. Commercial investment is now global. Private sector motivations for R&D are nearer-term and focus on return on investment. Semantic Wave 2008 predicts that both consumer-internet and enterprise-oriented investments in semantic technology will increase significantly through the end of the decade.

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Below:
What Are Semantic R&D Trends?



Source: Project10X

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Market Trend 2—Information & Communication Technology

ICT semantic technology markets will exceed \$50 billion in 2010.

The global ICT market is \$3.5 Trillion and will be \$4.3 Trillion by 2010. Growth in the E7 countries (China, Brazil, Korea, India, Russia, Mexico, and Taiwan) is currently around 20-percent per year. The market for semantic technologies is currently a tiny fraction of global ICT spending. But, growth is accelerating.

Semantic Wave 2008 profiles more than 270 companies that provide semantic technology R&D, services and products. Most are small, boutique firms, or start-ups. But, a significant number of established ICT companies have entered the semantic space. Overall, we estimate markets for semantic technologies in ICT exceed \$50 B in 2010.

Below:

Who Are the Semantic Technology Suppliers?

42 Objects	Contextware	H5	Lexxe	OpenLink Software	Sirma Group — Ontotext
Above All Software	Contivo	hakia	Liminal Systems	Open Text	Smart Desktop
Abrevity	Convera	HBS Consulting	Linguamatics	Oracle	SmartLogic
Access Innovations	Copernic	Hewlett-Packard	Linguistic Agents	PhraseTrain	Soar Technology
Active Navigation	Correlate	i2	LinkSpace	Polymeta	Software AG
Adaptive Blue	Cougaar Software	IAC Search & Media	Lockheed Martin	Powerset	Sony
Adobe Systems	Coveo	IBM	LogicLibrary	Pragatic	Spock
Aduna	Crystal Semantics	ILOG	Lymba Corporation	Profium	SRA International
Agent Logic	CureHunter	Image Matters	Magenta Technology	Progress Software	SRI International
Agent Software	Cycorp	Imindi	Makna Semantic Wiki	Project10X	Sun Microsystems
Agilense	Dassault Systemes	iMorph	Mandriva	Proximic	SunGard
Altova	Data-Grid	Infolution	Mark Logic	PTC	Sybase
Amblit Technologies	Day Software	Informatica	MatchMine	Quigo	Synomos
Apelon	Deepa Mehta	Information Extraction Sys.	McDonald Bradley	Radar Networks	SYS Technologies
Arisem	Design Power	InforSense	Mendix	Raytheon	System One
Articulate Software	DERI	InfoSys	MetaCarta	Readware	TACIT
AskMe	Design Power	Innodata Isogen	MetalIntegration	Rearden Commerce	Talis
AskMeNow	DFKI	Intellidimension	Metallect	Recommind	Taxonomy Strategies
Aspasia	DiCom Group	Intelligent Automation	Metatomix	Red Hat	TÉMIS Group
Astoria Software	Digital Harbor	Intellisemantic	Metaview 360	Reengineering	Teradata
AT&T Research	Digital Reasoning Sys.	Intellisophic	MetaWeb Technologies	Reinvent Technology	Teragram
ATG	Discovery Machine	Interwoven	Métier	Revelytix	TextDigger
Attensity	DreamFactory Software	Invention Machine	Microsoft Corporation	RuleBurst	Textual Analytics
Autonomy	EasyAsk	Iona Technologies	Mind-Alliance Systems	SAIC	Textwise
Axontologic	Effective Soft	Irion Technologies	Mindful Data	SaltLux	The Brain Technologies
BAE Systems	Ektron	Iron Mountain	Miosoft	Sandpiper Software	The METADATA Co.
BBN Technologies	EMC Corporation	iSOCO	Modulant	SAP	Thetus
Be Informed	Empolis	ISYS Search Software	Modus Operandi	SAS Institute	Thinkmap
BEA Systems	ENDECA	Janya	Molecular	SchemaLogic	Thomson Corporation
Biowisdom	Enigmatic	JARG Corporation	Mondeca	Semandex Networks	ThoughtExpress
Boeing Phantom Works	Enterra Solutions	Joost	Moresophy	Semansys Technologies	TopQuadrant
Bouvet	Entrieva	JustSystems	Motorola Labs	Semantic Arts	Triplt
Bravo Solution	Epistemics	K2	mSpace	Semantic Discovery	Troux Technologies
Business Semantics	Expert System	Kalido	Nervana	Semantic Edge	True Knowledge
Cambridge Semantics	ExpertMaker	Kapow Technologies	Netbreeze	Semantic Insights	Ultimus
Celcorp	Factiva	Kennen Technologies	Netezza	Semantic IQ	Ultralingua
Celtx	Fair Isaac	Kirix	NetMap Analytics	Semantic Knowledge	Versatile Info Systems
CheckMI	Fast Search & Transfer	Knewco	NeurokSoft	Semantic Light	Vignette
Cisco Systems	Fortent	Knova Software	Nielsen BuzzMetrics	Semantic Research	Vitria
Clarabridge	Fortius One	Knowledge Based Sys.	Noetix	Semantic Search	Vivisimo
ClearForest	FourthCodex	Knowledge Computing	Nokia	Semantic Solutions	Vivomind Intelligence
Cogito	Franz Inc.	Knowledge Concepts	Northrop Grumman	Semantic System	WAND
CogniIT a.s	Fujitsu Laboratories	Knowledge Foundations	Novamente	SemanitiNet	WebLayers
Cognition Technologies	General Dynamics IT	Knowledge Media Inst.	nStein	Semantra	WiredReach
Cognium Systems	Generate	Knowledge Systems, AI Lab	NuTech Solutions	Semaview	Wordmap
Cohereweb	GeoReference Online	Kroll Ontrack	Ontology Online	SemperWiki	XSB
Collarity	Global 360	Kyiel	Ontology Works	SERENA Software	Yahoo!
Collexis	Google	Language and Computing	Ontomantics	SiberLogic	Zepheira
Composite Software, Inc.	Graphisoft	Language Computer Corp.	ontoprise	Siderean Software	ZoomInfo
Computas AS	Groxis	LEGO Americas	Ontos	Sierra Nevada Corp	Zotero
Computer Associates	Gruppometa	Leximancer	OntoSolutions	SilkRoad Technology	ZyLAB
Connotate					
Content Analyst					

Source: Project10X

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Consumer content, entertainment & advertising dollars will build Web 3.0.

and advertising as well as knowledge exchange and business efficiency. It is growing rapidly and it is taking market share (i.e., money) away from other media.

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Market Trend 4—Enterprise Horizontal

Middleware, services, processes, search, and collaboration go semantic.

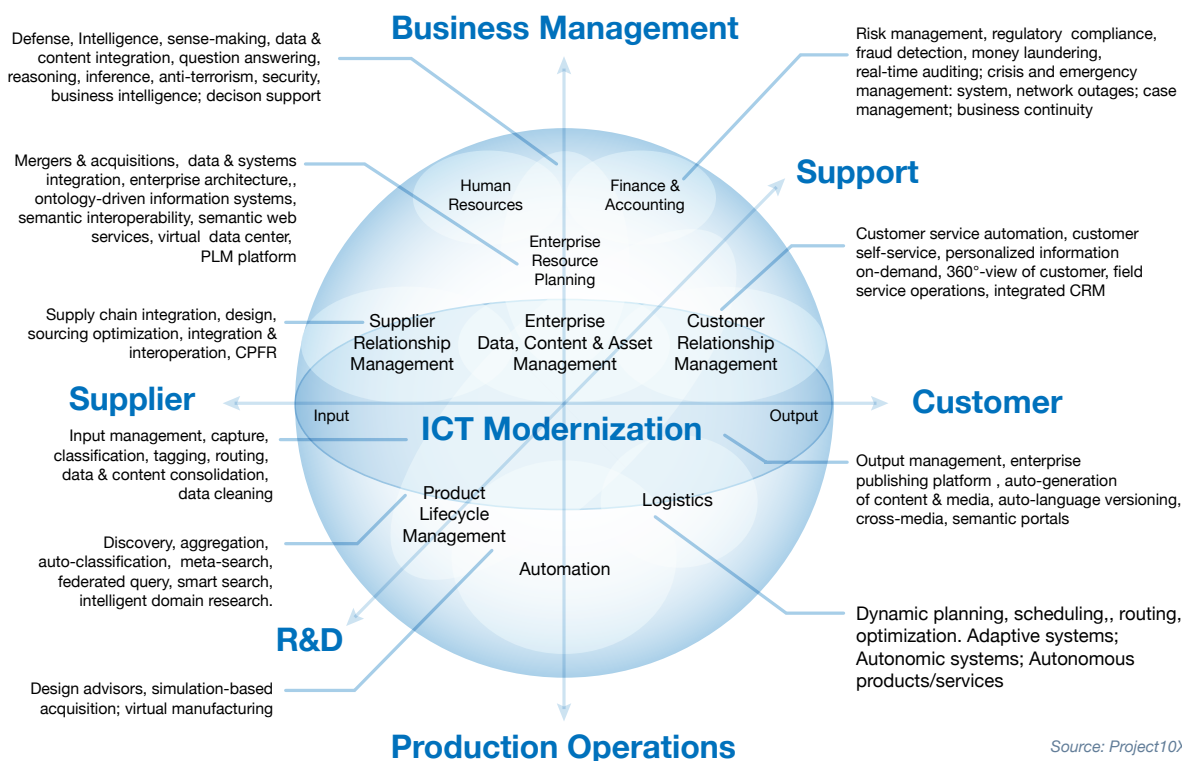
As shown in the diagram below, just about everywhere one can look in an enterprise, someone somewhere is applying semantic technologies to some problem. Drivers of enterprise business value are all strong— new capability, life cycle ROI, performance, and strategic edge.

Semantic Wave 2008 examines enterprise markets for semantic technologies including twelve categories of commercial-off-the-shelf software (COTS) packages that are estimated to represent a combined software product and service revenue of more than \$160 billion in 2010. The report projects the transition of these market segments from conventional to semantic COTS technologies. Amongst the first tier of large ICT technology providers, areas that are being targeted first are related to the internal stack, or plumbing for suites of applications because these changes make few

demands on customer while establishing a semantic application framework that developer can use as a foundation. Service oriented architecture becomes semantic SOA. Changes that impact application concept of operations, and user interface come next.

Also, enterprise software has a long tail. There are an estimated 56 million firms worldwide, including 1.5 million with more than 100 employees, and around 80,000 businesses with more than 1,000 employees. The transition to semantic software technologies will facilitate mass customization of commercial-off-the-shelf solutions enabling software vendors to address more levels of the market with sustainable solutions.

Below:
What Semantic Technologies Are Being
Employed in Enterprise?



Source: Project10X

Market Trend 5—Industry Verticals

150 case studies make the case that semantic wave markets are here.

Semantic Wave 2008 examines semantic technology adoption in industry verticals. The report summarizes 150 case examples in fourteen industry sectors. The table below highlights some of the semantic application case examples in eight vertical industry sectors. Each industry has both horizontal

and vertical needs. Applications are diverse. Nearly three-fourths of the case examples come from private industry. A little more than one-fourth are public sector. Collectively, they make a strong case that semantic wave markets are here and now.

Below:
What Are Industry Vertical Markets
for Semantic Technologies?

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MARKET SECTOR	EARLY ADOPTER ORGANIZATIONS	SEMANTIC APPLICATION CASE EXAMPLES
ADVERTISING, CONTENT & ENTERTAINMENT	BBC; Bertelsmann; Dentsu; Disney; Elsevier; Associated Press; NZZ	Digital asset management; rich media interoperability; content mining; mapping of concepts across content libraries; accelerated creation of new derivative information products; identification and extraction of information types, such as chemical compounds and classes for science; rapid development of custom news feeds; skills curation and collaboration
EDUCATION & TRAINING	Industry; universities; governments; ETH Zurich	E-learning; simulation learning tools ("learning by doing"); semantic collaboration environments; digital library services; rapidly customized coursework content; automated scoring; publication streamlining
ENERGY & UTILITIES	Air Liquide America; Air Products; BP; GE Infrastructure Water & Process Technologies; Shell Oil; Statoil	Energy exploration; processing real-time remote sensor data; power distribution through "common information models"; multi-agent technologies; corporate portals across departments and disciplines; adaptive data warehousing; multi-format document access; knowledge-based service reporting; proposal management; integrating information across operating units; product and market segmentation; scenario validation
FINANCIAL SERVICES	Citigroup; Ameriprise Financial; Aon; Fireman's Fund Insurance; UBS; Credit-Suisse; Swiss Re; Munich Re; Bank Vontobel	Risk and compliance management; due diligence; security and surveillance; analytical dashboards and composite applications; case management; auditing transparency; trend analysis; regulation and policy management; document and contract analysis; business rules for investment strategies; sales and customer service; risk scoring; new business acquisition; policy-based computing and application monitoring; loan processing; analyst productivity suites
HEALTH, MEDICINE, PHARMA & LIFE SCIENCES	National Library of Medicine; Amgen; Biogen; Eli Lilly; GSK; Novartis; Pfizer; Healthline; Partners Clinical Informatics; University of Texas; Mayo Clinic; ImpactRX; Cleveland Clinic; AstraZeneca	Meta-searching and clustering; enterprise search; scientific discovery; translational medicine; clinical knowledge bases; reasoning and decision support; healthcare supply chain planning; in silico drug discovery; integrated biosurveillance; lexical standardization; market intelligence; patient records; drug development cost reduction
MANUFACTURING	Emerson Motors; General Dynamics; General Motors; BAE Systems; Rockwell Automation; Proctor & Gamble; EniTechnologies; Siemens	R&D; supply chain; customer support; product modeling; design and fabrication; design-to-order; document life cycle management; virtual manufacturing; international market and scenario simulation and visualization; robotics and autonomous systems; speech recognition; automobile telematics and automation; quality improvement; enterprise knowledge management; inventory optimization; maintenance and repair management; competitive intelligence; intellectual capital management; portfolio management; customer self-service
TRANSPORTATION & LOGISTICS	Tankers International; SouthWest Airlines	Cargo management; shipment tracking; contract review & management; logistics outsourcing management; logistics cycle emulation; network routing and scheduling
PUBLIC SECTOR	DoD Finance & Accounting Service; GSA; National Communications System Continuity Communications Working Group; FAA; OMB; National Geospatial Intelligence Agency; National Institutes of Health; National Cancer Institute; National Center for Biomedical Ontology; Dept. of Health and Human Services; Defense Information Systems Agency; Defense Logistics Agency; U.S. Army; XVIII Airborne Corps; Dept. of Education; Internal Revenue Service; National Biological Information Infrastructure; NSA; CIA; DIA; Dept. of Homeland Security	Semantic Service Oriented Architecture (SSOA); modeling IT environments; federated queries across databases; process management; geospatial information interoperability; predictive analytics; document parsing and entity extraction; mapping biological networks and biomarkers; unified medical ontologies; clinical care support; net-centric data services; knowledge navigation; speed of command; combat information distribution; nested networks; educational and training gateways; grant application processing; tax code navigation; expert systems; integrated defense information access; relationship analytics and social network analysis; pattern recognition; emergency management; immigration; infrastructure protection; international trade

About the Author



Mills Davis is founder and managing director of Project10X — a Washington, DC based research consultancy specializing in next wave semantic technologies and solutions. The firm's clients include technology manufacturers, global 2000 corporations, and government agencies.

Mills served as principal investigator for the Semantic Wave 2008 research program. A noted consultant and industry analyst, he has authored more than 100 reports, whitepapers, articles, and industry studies.

Mills is active in both government and industry-wide technology initiatives that are advancing semantic technologies. He serves as co-chair of the Federal Semantic Interoperability Community of Practice (SICoP). Mills is a founding member of the AIIM interoperable enterprise content management (iECM) working group, and a founding member of the National Center for Ontology Research (NCOR). Also, he serves on the advisory boards of several new ventures in the semantic space.

In addition to his research and consulting practice, Mills is currently directing development of community based collaborative semantic magazine that is dedicated to aggregating, linking, and making sense of all things Web 3.0.

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Above:
Sample Pages from [Semantic Wave 2008: Industry Roadmap to Web 3.0](#)

Summary

In the preceding pages we introduced the thesis and have highlighted some findings and conclusions from our new research report — *Semantic Wave 2008: Industry Roadmap to Web 3.0*. We hope this brief overview will encourage you to read the full report. As you can see from the sample pages, Semantic Wave 2008 is no ordinary research report. It is written to be understood by a broad audience and contains a great many figures and illustrations.

Semantic Wave 2008 explains the new semantic technology and gives perspective on emerging patterns and keys to success. It gauges both technology and market readiness. By mapping the frontier, it shows where the tough problems are, and where to look for breakthroughs. But, most importantly, Semantic Wave 2008 profiles significant opportunities for executives, developers, designers, entrepreneurs, and investors. What to build and what to buy, and why. For this, SW2008 is simply the most comprehensive resource available anywhere at this crucial time.

The technology section of the report examines five strategic technology themes and shows how innovations in these areas are driving development of new categories of products, services, and solution capabilities. Themes include: executable knowledge, semantic user experience, semantic social computing, semantic applications, and semantic infrastructure. The study examines the role of semantic technologies in more than 100 application categories. An addendum to the report surveys more than 270 companies that are researching and developing semantic technology products and services.

The market section of the report examines the growth of supply and demand for products, services and solutions based on semantic technologies. Specifically, the report segments and discusses semantic wave markets from five perspectives: research and development, information and communication technology, consumer internet, enterprise horizontal, and industry verticals. Viewed as horizontal and vertical market sectors, each presents multi-billion dollar opportunities in the near- to mid-term. The study presents 150 case studies in 14 horizontal and vertical sectors that illustrate the scope of current market adoption.

In addition to the main report, there are two addenda: a supplier directory, and an annotated bibliography.

Specifications for the Semantic Wave 2008 report and a topic outline follow this page.

Semantic Wave 2008: Industry Roadmap to Web 3.0

Report Specifications

Report Outline

Format	PDF — Color and B&W	1 Introduction
Pages	400	2 Semantic Wave
Figures	290	2.1 Strategic Vision
Vendors	270	2.2 Web 1.0
Applications	110	2.3 Web 2.0
Market sectors	14	2.4 Web 3.0
Case examples	150	2.5 Web 4.0
Price	\$3495 USD	3 Semantic Technologies
Availability	Nov 1, 2007	3.1 Technology Themes & Perspectives
		3.2 Knowledge
		3.3 Semantic User Experience
		3.4 Semantic Social Computing
		3.5 Semantic Applications
		3.6 Semantic Infrastructure
		4 Semantic Markets
		4.1 Market View
		4.2 Research and Development
		4.3 Information & Communications Technology
		4.4 Consumer Internet
		4.5 Enterprise Horizontal
		4.6 Industry Verticals
		Addenda
		A Suppliers
		B Bibliography

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URL: www.project10X.com

For additional information, please contact:

*Karen Aiken, 408-234-9054 or
Bojana Fazarinc, 408-334-4708*

Semantic Exchange

Semantic Exchange is a collaborative industry news, research, and education initiative about all things web 3.0 and semantic web. This program is sponsored the industry leading organizations presented here. The Semantic Exchange website is part community wiki, part internet magazine, part semantic technology sandbox, and part knowledge outfitter.



Be Informed

A leading supplier of semantic software and infrastructure for knowledge management. Semantic framework integrates subject and process ontologies with business rules to deliver intelligent e-forms, knowledge bases over heterogeneous information sources, advisory services, auto-classification, decision trees, calculators, and semantic search and navigation.



Celtx

Celtx is the world's first fully integrated solution for media pre-production and collaboration powered by semantic technologies. This engaging, standards based software for the production of film, video, theatre, animation, radio and new media, replaces old fashioned 'paper, pen & binder' media creation with a digital approach to writing and organizing that's more complete, simpler to work with, and easier to share.



CHECKMi

CHECKMi semantic solutions improve the control, agility and cost of service oriented agent & semantic grid computing. The CHECKMi: Mate is a product platform for networking semantic software agents together to power information analytic services and deliver secure business processing.



Cycorp

Developer of large-scale "ontology of the universe," common sense knowledge base, and associated reasoning systems for knowledge-intensive applications. Cyc KB provides a deep layer of understanding that is divided into thousands of "microtheories", each of which is essentially a bundle of assertions that share a common set of assumptions about a particular domain of knowledge, a particular level of detail, a particular interval in time, etc.



Digital Harbor

The industry-leading Operational Intelligence Software Suite for proactive management of risk and compliance. Serving public sector, retail and healthcare Industries our operational risk and compliance solutions include theft, fraud & criminal Investigations, improper payments, intelligence fusion, emergency management and auditory compliance. This platform delivers real-time analytics, case management and dynamic dashboard technologies for detection, investigation, assessment and monitoring.



empolis

empolis, The Information Logistics Company, offers enterprise content and knowledge management solutions for company-wide information logistics and for improving business processes. empolis' core competencies are information management, service management, product & catalog management and media management. empolis consistently relies on open standards, such as XML, Java or OWL and RDF.



Expert System is the leading provider of semantic software, which discovers, classifies and interprets text information. The company develops a patent pending technology, Cogito, that enables organizations to: extract, discover and understand the connections in your strategic information sets – the thousands of files, e-mails, articles, reports, web pages you have access to everyday; and, understand automatically the meaning of any text written in the language we use to communicate (natural language). Cogito improves business decisions in real time for the majority of corporate functions.



KFI

KFI's Mark 3 Associative Knowledge Platform enables cost effective, long-term retention and application of enterprise knowledge, providing intelligent business process, integrated financial simulations, knowledge-based training, and complex decision support.



Metatomix

The Metatomix Semantic Platform integrates data, uncovers and defines information relationships, and provides meaning and actionable insight to applications. It does so by creating a real-time virtual integration layer that non-invasively accesses data from any source (static and dynamic) and allows it to be understood & leveraged by practically any application.

Whether you are a newcomer to semantic technologies or already have experience with them, the goal of Semantic Exchange is to help you better keep up with the rapid pace of technology and infrastructure development, connect with the people and companies making the next stage of the internet happen, and understand the breadth of applications across consumer and enterprise industry sectors.



Mondeca

Mondeca provides software solutions that leverage semantics to help organizations obtain maximum return from their accumulated knowledge, content and software applications. Its solutions are used by publishing, media, industry, tourism, sustainable development and government customers worldwide.



Netbreeze

Leading European supplier of knowledge generation solutions that use semantic technologies and artificial intelligence to extract knowledge from varied Internet sources and integrate it with business processes. Applications include early warning, risk management, marketing and sales tools, media monitoring, asset management, corporate governance and compliance, issue management, anti money laundering, know-your-client, executive search, project management, and other business applications.



Ontotext

Ontotext is a semantic technology lab of Sirma Group. Ontotext researches and develops core technology for knowledge discovery, management, and engineering, Semantic Web, and web services.



Project10X

Project10X is a premier industry research, education, and consulting firm specializing in next wave semantic technologies, solutions, markets, and business models. Project10X publishes the Semantic Wave research series including the Semantic Wave 2008 Report. The firm provides educational and training services, and consults with technology manufacturers, global 2000 corporations, government agencies, and technology start-ups. Project10X is directing the Semantic Exchange industry education initiative.



Radar Networks

Twine is the first application on the Radar Networks Semantic Web platform. Twine helps users leverage and contribute to the collective intelligence of their friends, colleagues, groups and teams



Sandpiper

Develops W3C and OMG standards-compliant, semantically aware, knowledge-based software products that facilitate business information interoperability, terminology normalization and context resolution across web-based and enterprise information systems. Visual Ontology Modeler™ (VOM) UML-based ontology modeling environment supports frame-based knowledge representation and construction of component-based ontologies that capture and represent concepts, resources and processes.



Semantic Arts

Semantic Arts is a USA-based consulting firm that helps large firms transform their Enterprise Architectures. Our specialty is reducing complexity through the intelligent use of Semantic Technology and Service Oriented Architecture.



Semantic Insights

Leading supplier of semantic research solutions. SI services help people access the internet to read just the information they're interested, use their computer to help reason about it, and then report it just the way they want, easy, fast, and automatically.



Semantic System

Semantic System ag manufactures hardware technology for intelligent computer systems. Its first generation computer chip "thinks" like a biologic brain making it possible to run complex thought and analyzing processes in hardware to obtain results equivalent to those obtained manually by a skilled humans.



Semantic Universe

Semantic Universe's mission is to raise awareness and explain the usage of semantic technologies in business and consumer settings. Projects by Semantic Universe include the annual Semantic Technology Conference (HYPERLINK "<http://www.semantic-conference.com/>"www.semantic-conference.com) and the SemanticReport newsletter (HYPERLINK "<http://www.semanticreport.com/>"www.semanticreport.com).



Smartlogic

Ontology driven technology solutions for information management projects. Professional services and semantic middleware for government, media, and financial services industries. Semaphore semantic processor provides description logic based automatic classification and categorization for taxonomies, thesauri, and ontologies based on rules., intelligent guided search, taxonomy management, dynamic profiling and recommendation software for intranet, internet and portal applications.



Talis

The Talis Platform is an open technology platform for mass collaboration and human-centric and information-rich applications. It combines Semantic Web, information retrieval, collective intelligence, and behavioral mining technologies, which can be accessed through a suite of RESTful web services. Talis Platform provides data management, organization and analysis components that can learn and understand patterns of behavior and present them through an API to be interwoven into applications.



Textwise

Leading supplier of fully automated and real-time contextual targeting services for both advertisements and web pages. Natural Language Processing (NLP) and Machine Learning (ML) technologies automate establishment of semantic signatures with contextual attributes that enable high precision customer targeting and media placement.



TopQuadrant

TopQuadrant provides products, services, knowledge, training programs and methods to help organizations integrate data and processes and to harness the knowledge distributed across systems and parties. TopQuadrant helps customers implement new capabilities for integration, policy management, search, enterprise architecture and model-driven applications. The TopBraid product suite provides an enterprise-level platform for developing and deploying semantic applications.



WAND

WAND provides structured multi-lingual vocabularies with related tools and services to power precision search and classification applications on the internet, including custom travel, jobs and skills, and medical taxonomies to our cornerstone and product and service taxonomies. In addition to licensing its taxonomies for integration into third party applications, WAND builds precision online horizontal and vertical business directory applications.